

Data Visualization

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Year 2017
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Short Bio of Phuc Do

- Phuc Do is an Associate Professor of Information Systems at the University of Information Technology, Vietnam National University of Ho Chi Minh City.
- He currently works mostly on knowledge discovery technologies and their application in social network analysis by using topic modeling .
- Phuc Do has written about 80 research papers and books. He has been the Principal Investigator of several projects in Knowledge Discovery From Data, Text Analysis and Summarization, Topic Based Social Network Analysis, Citation Network Analysis.

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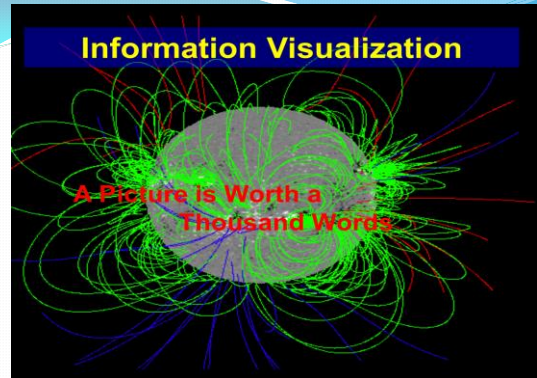
Outline

- Data Visualization
- Examples
- Tree Visualization
- Graph Visualization
- Text Visualization

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Information Visualization



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Visualization & Information Visualization:

- Visualization:
 - "The use of computer-supported, interactive, visual representations of data to amplify cognition."
 - Goal: discovery, decision making, explanation
- Information Visualization:
 - "The use of **computer-supported?**, interactive, visual representations of abstract data to amplify cognition."

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Visualization Amplifies Cognition

1. Increase memory and processing resources
2. Reduce information searching
3. Enhance detection of patterns
4. Enable perceptual inference operations
5. Use perception attention for monitoring
6. Encode information in a manipulative medium

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An example of Data Mining Visualization: Association rules

List of Associations

[illegible]

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Drawbacks

- Long list of rules
- Many rules are redundant
- Very hard to capture the important rules
- How to access association rules effectively

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Graph of Association rule

- Association rules $X \rightarrow Y$
- Edge
 - Source: X
 - Target: Y
- Set of Association rules
 - Graph (V,E)
 - V: set of frequent itemsets
 - E: set of edges created from association rules

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Graph created from association rules

[illegible]

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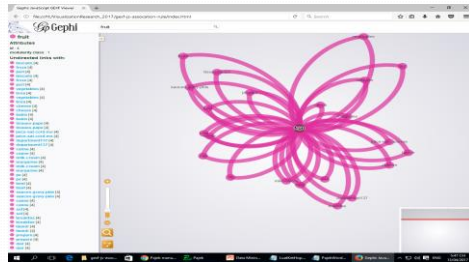
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Graph visualization by Pajek software

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Graph Visualization by Gephi on the Web



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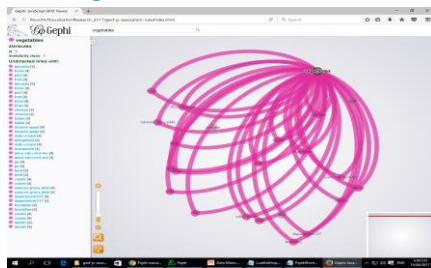
Interactive mode in Graph visualization

- Very easy to capture the important association rules
- Click on node to discover the co-occurrence items
- Move around to discover the association rules

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Discover the items co occurrence with vegetables



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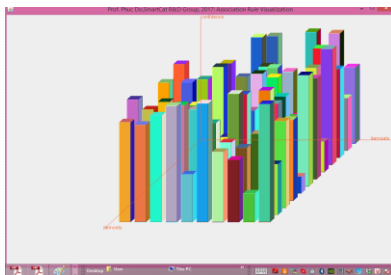
Using cube to visualize AR

- Each rule is a cube
- Click on Cube to visualize the association rule
- The height of Cube: the confidence of AR.
- Easy to discover the important rules based on the height of cube.

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AR visualization, each AR is a cube



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Visualization components

- Color
- Size
- Texture
- Proximity
- Annotation
- Interactivity
 - Selection / Filtering
 - Zoom
 - Animation

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Visualization of Relations

- Tree Visualization
- Graph Visualization

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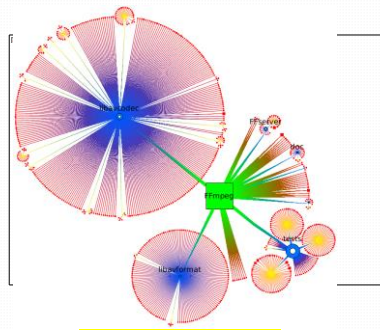
Tree Visualization

- Treemap method: visualize the tree structure that use virtually every pixel of the display space to convey information
- Every subtree is represented by a rectangle, that is partitioned into smaller rectangles with correspond to its children.
- The position of the slicing lines determines the relative sizes of the child rectangles.
- For every child, repeat the slicing recursively, swapping the slicing direction from vertical to horizontal or conversely

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Tree Visualization



Bubble-Tree Layout
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Tree Visualization



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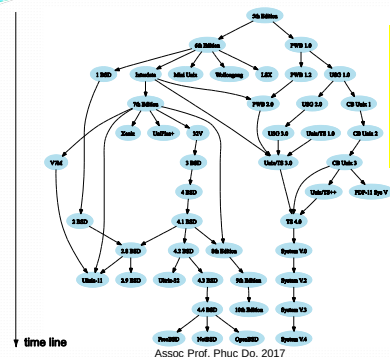
Graph Visualization

- Graphs are the most general type of relational data
- Similar to a tree, it consists of nodes and edges
- Different from a tree, a child node may have multiple parent nodes
- A graph contains loops, or multiple paths between two nodes in the graph.

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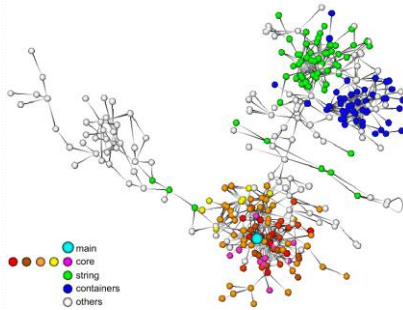
Graph Visualization



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Graph Visualization



Call Graph using a Force-Directed Layout

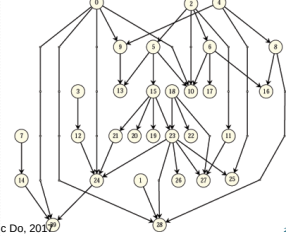
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Layout of Directed Graphs

- Layering (<http://www.csus.yk.ue/staff/NikolaNikolov/#phd>)
- Algorithm to calculate the co-ordinate of vertices

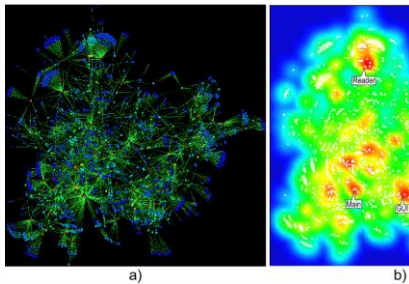
Crossings: 11



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Graph Visualization



a)

b)

Force-Directed Layout

Splatting;
Dense Representation

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Text Visualization and Knowledge Discovery for Text

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Text Visualization and Knowledge Discovery for Text

- **Abstract**
- Text is one of the greatest data and popular around us. Text contains our history and is a major mean to recording information and knowledge. With the Internet, text data has been generated with significantly speed. Currently, millions of websites are generating extraordinary amount of online text data everyday. The users of the social network like FaceBook, Twitter,... produce billions of posting messages everyday.
- The explosion of the data makes a lot of challenges to discover and understand the knowledge in text.
- Text visualization techniques can be helpful for solving these problems. Various visualizations have been designed for showing the similarity of text documents, revealing and summarizing text content, and helping discover the big text data exploration.
- In this talk, we will show the techniques of text visualization and the application of text visualization in knowledge discovery for text. We also address the challenges for text visualization with big text data in big data era.

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Why visualize text?

- **Understanding** – read a document
- **Summaries** – get the “key” topic of a document
- **Clustering** – group together similar contents
- **Correlate** – compare patterns in text to those in other data, e.g., correlate with social network

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Summarize with Words

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Words are (not) nominal?

High dimensional

- High dimensional (10,000+)
- Words have meanings and relations
 - Correlations: *Hong Kong, San Francisco, Bay Area*
 - Order: *April, February, January, June, March, May*
 - Hierarchy, antonyms & synonyms, entities, ...(ontology)

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Tips: Tokenization and Stemming

- Tokennizer, Vntokennizer
 - "Thuyền ai thấp_thoảng bên sông",
 - "Đưa câu mái đây chạnh_lòng nước non"
 (Hue Fork Song)
- Stemming
- Name Entity Recognition

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Bag of Words and Vector Model

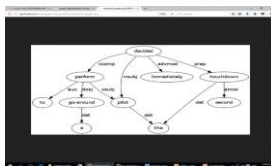
- Ignore ordering relationships within the text
- A document $\approx$ vector of term weights
 - Each dimension corresponds to a term (10,000+)
 - Each value represents the relevance
 - For example, simple term counts
- Aggregate into a document-term matrix
 - Document vector space model
 - Frequency of word in document

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Graph based text representation

- Frequent words:
Vertices
- Link: the co occurrence words in sentence.
- Co occurrence graph
- Preserve the order of words in sentence



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Word Clouds

- **Strengths**
 - Can help with initial idea about text
- **Weaknesses**
 - Term frequency may not be meaningful
 - Does not show the structure of the text

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Huế Folk-song

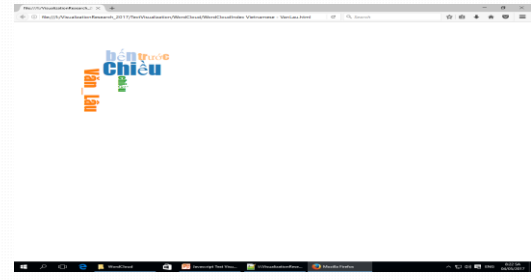
- "Chiều chiều trước bến Văn_Lâu",
- "Ai ngồi ai câu ai sầu ai thảm",
- "Ai thương ai cảm ai nhớ ai trông",
- "Thuyền ai thấp_thoảng bên sông",
- "Đưa câu mái_dây chạnh_lòng nước non"



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Word Cloud



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Word Cloud



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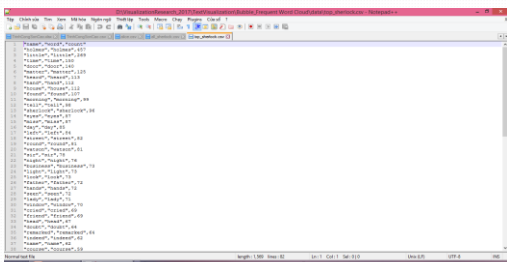
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Bubble Cloud Visualization of frequent words

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Frequent word list



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Bubble Cloud



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Figure 1. The proposed system architecture.

SE

Đoàn Văn Phúc, Đỗ Văn

Latent Dirichlet Allocation

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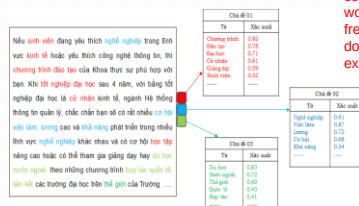
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Latent Dirichlet Allocation (LDA)

- Invented by Prof. David Blei, 2003
- Probabilistic Graphical Model (PGM)
- Generative probabilistic modeling
 - Treats data as observations
 - Contains hidden variables
 - Hidden variables reflect thematic structure of the collection.
- Infer hidden structure using posterior inference
 - Discovering topics in the collection.

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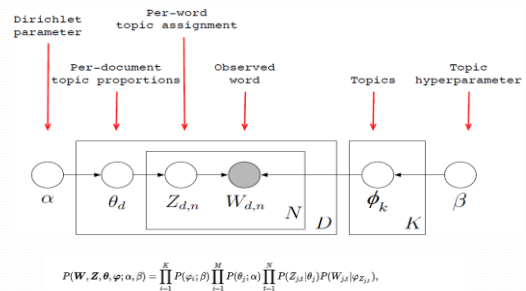
Using LDA to discover the latent topics of messages



LDA is a generative probabilistic model of a corpus. The basic idea is that the documents are represented as random mixtures over latent topics, where a topic is characterized by a distribution over words.

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LDA as Graphical Model



$$P(\mathbf{W}, \mathbf{Z}, \boldsymbol{\theta}, \boldsymbol{\varphi}; \alpha, \beta) = \prod_{d=1}^D P(\boldsymbol{\varphi}; \beta) \prod_{n=1}^N P(z_{d,n}; \alpha) \prod_{n=1}^N P(w_{d,n} | z_{d,n})$$

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Generating Process of LDA

- Choose $N \sim \text{Poisson}(\xi)$
- Choose $\boldsymbol{\theta} \sim \text{Dir}(\alpha)$
- For each of the N words w_n
 - Choose a topic $z_n \sim \text{Multinomial}(\boldsymbol{\theta})$
 - Choose a word w_n from $p(w_n | z_n, \beta)$ a multinomial probability conditioned on the topic z_n β is a k by n matrix parameterized with the word probabilities.

$$[\beta]_{k \times V} \quad \beta_{ij} = p(w^j = i | z^i = 1)$$

Inference and Parameter Estimation

- Parameters:
 - Corpus level: α, β : sampled once in the corpus creating generative model.
 - Document level: $\boldsymbol{\theta}, \mathbf{z}$: sampled once to generate a document
- Inference: estimation of document-level parameters.

- However, it is intractable to compute $p(\boldsymbol{\theta}, \mathbf{z} | \mathbf{w}, \alpha, \beta) = \frac{p(\boldsymbol{\theta}, \mathbf{z}, \mathbf{w} | \alpha, \beta)}{p(\mathbf{w} | \alpha, \beta)}$, which needs approximate inference.

Gibbs Sampling for LDA

- Gibbs Sampling can be used to sample ϕ and θ . They are calculated by.

$$\hat{\phi}_k^w = \frac{N_k^w + \beta}{N_k^{(\cdot)} + W\beta} \quad \hat{\theta}_k^d = \frac{N_k^d + \alpha}{N_{\cdot}^d + K\alpha}$$

ϕ is a probability dist. of **topic x word**
 θ is a probability dist. of **doc x topic**

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Latent topics without label

Topic 4		Topic 12		Topic 17	
đain	0.03189	phòng	0.03925	công ty	0.03245
hỏi	0.01957	quản lý	0.02322	việc làm	0.01767
Bác	0.01769	tin chi	0.01933	kiến tập	0.01485
học_dùng	0.01689	học phí	0.01402	nhận_vấn	0.01093
đain_vấn	0.01433	giảng dạy	0.01021	tham quan	0.00952
nội dung	0.01277	thi	0.00967	lương	0.00711
công tác	0.00677	học, lai	0.00833	nguồn	0.00429
đain_học	0.00597	thời gian_học	0.00790	nhà trường	0.00441
quản_dinh	0.00535	phòng học	0.00718	học_dùng	0.00500
phòng trọ	0.00499	máy chiếu	0.00554	học hỏi	0.00359

Topic 2		Topic 7		Topic 14	
chức lương	0.04213	cầu	0.06212	nguồn gốc	0.04539
sách_vấn	0.02543	số	0.05423	học phí	0.03478
hỏi_thảo	0.01456	yêu cầu	0.04562	đại học	0.03941
soi_danh	0.01213	thời gian	0.03412	nhà trường	0.03415
kinh nghiệm	0.01122	thời gian	0.03421	học phí	0.02727
hỏi_đáp	0.00992	học, lai	0.03121	quản_dinh	0.01325
khóa học	0.00782	cảm giác	0.00872	nhà ở	0.00932
nguồn gốc	0.00612	thầy	0.00780	học bổng	0.00823
phòng trọ	0.00467	cổ	0.00632	nhà trường	0.00763
đại học	0.00421	quản_dinh	0.00432	đến ngoài	0.00349

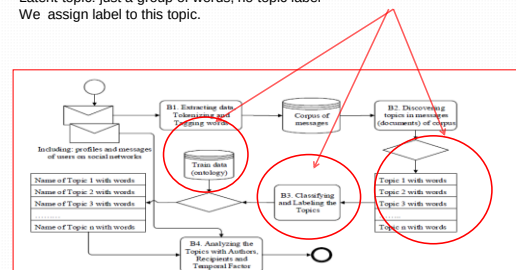
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Automatic labeling of topics

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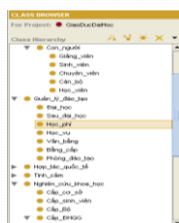
Automatic labeling by classification

Latent topic: just a group of words, no topic label
 We assign label to this topic.



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A topic taxonomy of university



We build a topic taxonomy of university
 Then we collect 200-300 messages for each topic
 We use LDA to discover the words representing for this topic
 SVM is used for classification two sets of words.

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Table 1. List of labeled topics and words distribution with probability

Topic 4 "social activity"		Topic 12 "management training"		Topic 17 "recruitment jobs"	
union	0.03189	Office	0.03925	company	0.03245
associate	0.01957	management	0.02322	job	0.01767
Uncle Ho	0.01769	credits	0.01933	fresher	0.01485
activity	0.01689	school fee	0.01402	staff	0.01093
union member	0.01433	teaching	0.01021	visit	0.00952
content	0.01277	test	0.00967	Salary	0.00711
mission	0.00677	learning	0.00833	people	0.00429
school year	0.00597	schedule	0.00790	environment	0.00441
regulation	0.00535	classroom	0.00718	activity	0.00500
movement	0.00499	projector	0.00554	learning	0.00359

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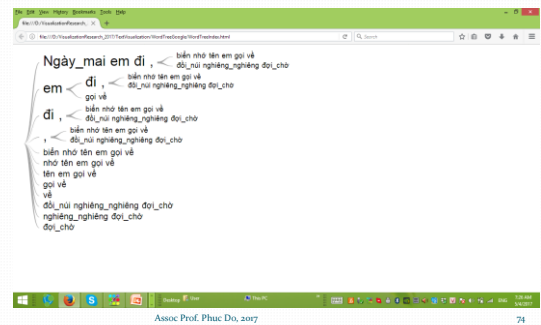
Biển nhớ, Trịnh Công Sơn

- "Ngày_mai em đi , biển nhớ tên em gọi về",
- "Ngày_mai em đi , đôi_núi nghiêng_nghiêng đợi_chờ",

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Word Tree



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Graph based document summarization

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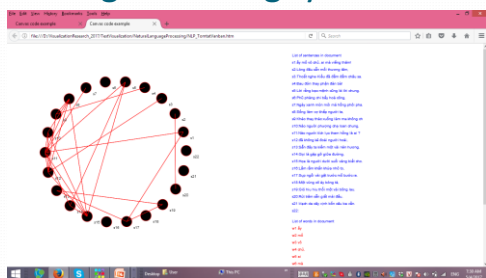
Text Visualization Summary

- High Dimensionality
 - Where possible use **text to represent text**...
 - ... which terms are the most descriptive?
- Context & Semantics
 - Provide **relevant context** to aid understanding.
 - Show (or provide access to) the **source text**.
 - Modeling Abstraction
- Determine your **analysis task**.
 - Understand abstraction of your **language models**.
 - Match analysis task with appropriate tools and models.

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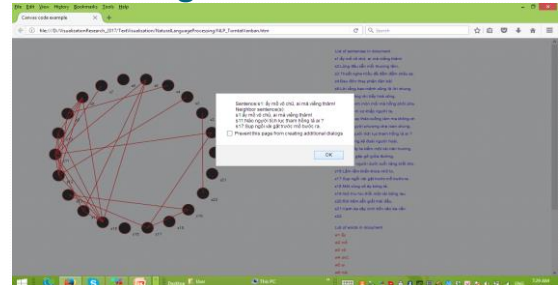
Kieu legend of Nguyễn Du



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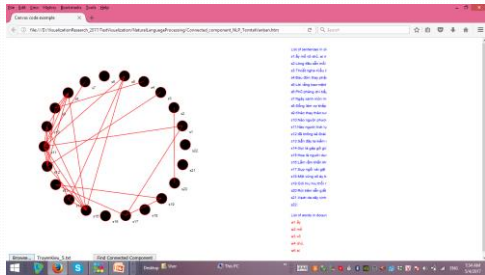
Clustering the sentences



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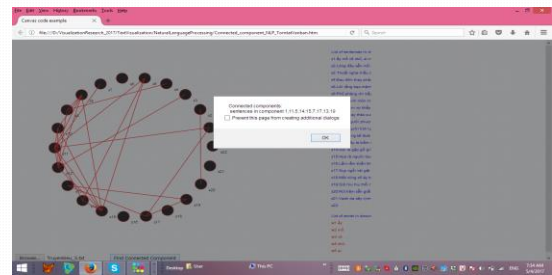
Connected Components



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Discover the connected component



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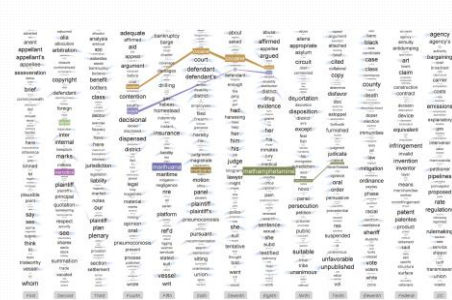
Doc. Similarity & Clustering

- In vector model, compute distance among docs
 - For TF.IDF, typically cosine distance
 - Similarity measure can be used to cluster
- Topic modeling approaches
 - Assume documents are a mixture of topics
 - Topics are (roughly) a set of co-occurring terms
 - Latent Semantic Analysis (LSA): reduce term matrix
 - Latent Dirichlet Allocation (LDA): statistical model

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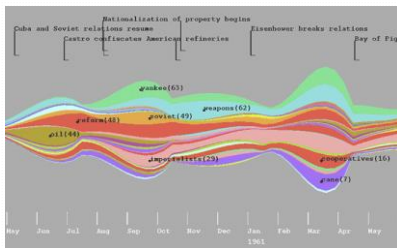
Parallel Tag Clouds [Collins et al 09]



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ThemeRiver [Havre et al 99]



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What topics they used in their email

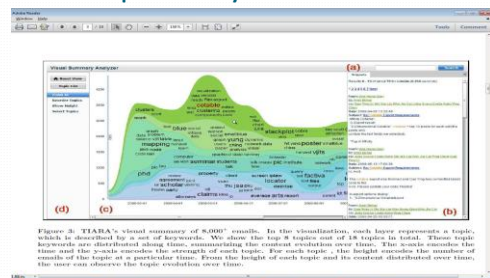


Figure 2: TIARA's visual summary of 8,000th results. In the visualization, each curve represents a topic, which is described by a set of keywords. We show the top 8 topics out of 18 topics in total. These topic keywords are distributed along time, corresponding the temporal evolution over time. The curve denotes the results of the topic at a particular time. From the height of each topic and its content distributed over time, the user can observe the topic evolution over time.

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Visualizing Sentiments and Emotions

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Sentiment analysis

- Sentiment analysis, also known as opinion mining, is one of the most important text mining tasks and has been widely used for analyzing, for example, reviews or social media data for various of applications, including marketing and customer service.
- The result of sentiment analysis is negative, neutral, positive

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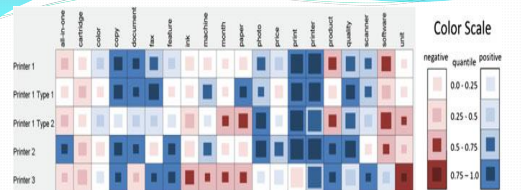
Sentiment indexing of Twitter data in a time-series diagram. This figure shows that the public sentiment regarding different events in real-life may change dramatically



C. Nan and W. Cui, *Introduction to Text Visualization*, Atlantis Briefs in Artificial Intelligence, DOI 10.2391/978-94-6239-860-4_6

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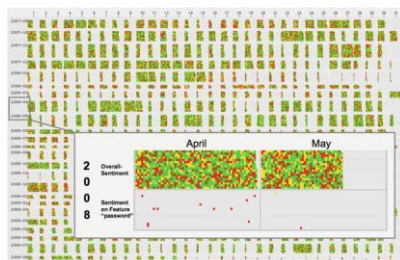
Summary report of printers: each row shows the attribute performances of a specific printer. Blue color represents comparatively positive user opinions and red color comparatively negative ones (see color scale). The size of an inner rectangle indicates the amount of customers that commented on an attribute. The larger the rectangle the more comments have been provided by the customers

C. Nan and W. Cui, *Introduction to Text Visualization*, Atlantis Briefs in Artificial Intelligence, DOI 10.2391/978-94-6239-860-4_6

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Pixel map calendar: each pixel indicates a document with the color encoding its overall sentiment, i.e., the average of all contained feature sentiments. Here, red indicates negative, green indicates positive, and yellow indicates neutral. In the background, the x-axis bins are days and y-axis bins are years with months



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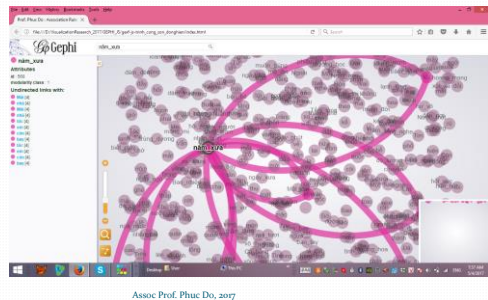
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Gephi Viewer

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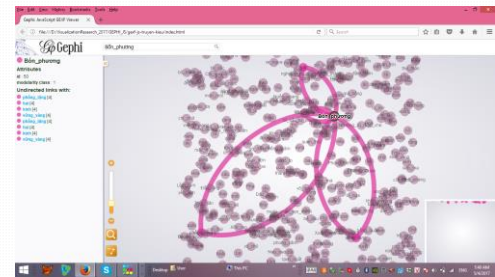
Trịnh Công Sơn Co-occurrence Word Graph



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Kieu Legend, Nguyễn Du Co-occurrence Word Graph



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Some Challenges in Text Visualization

- Interactive Visualization of Very Large Graphs
- Develop novel interactive visualizations that are tightly coupled with automatic natural language processing methods to enable human users to explore their data.
- Develop Real-Time Visualization of Streaming Text
- How interactive visualization can assist investigative analysis for large corpus

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Visualization software

- Host language (C/C++/Java/Python) plus OpenGL
- Stat/math package with graphics
 - R
 - MATLAB
- Special-purpose info visualization software
 - Earth mapping, biological network visualization, etc.
- Browser-enabled graphics/info visualization packages
 - [Google Charts](#)
 - [Processing](#) / [Processing.js](#)
 - [D3](#)
 - Java + Flash

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Hands-on

- [HTML intro](#)*
- [Google charts](#)
- [D3](#)

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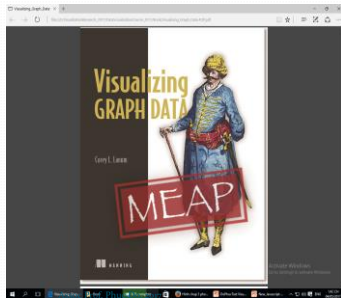
Resources

- Books
 - [Visual Complexity: Mapping Patterns of Information](#), Manuel Lima
 - [The Visual Display of Quantitative Information](#), Edward Tufte
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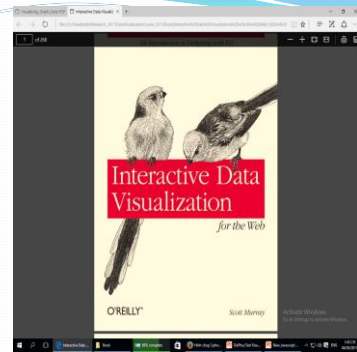
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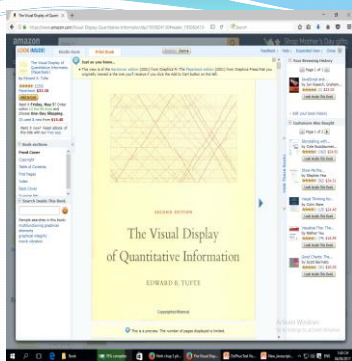


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Thank you for your attention!

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